

| Question |        |        |        | Marking details                                                                                                                                                                                                              |  |  |  | Marks available |        |        |        |        |   |       |  |       |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
|----------|--------|--------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|-----------------|--------|--------|--------|--------|---|-------|--|-------|--|------|---|---|--|--|--|--|--|---|--|--|--|--|--|--|--|
|          |        |        |        |                                                                                                                                                                                                                              |  |  |  | AO1             |        | AO2    |        | AO3    |   | Total |  | Maths |  | Prac |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
| 4        | (a)    | (i)    |        | Nucleus                                                                                                                                                                                                                      |  |  |  | 1               |        |        |        |        |   | 1     |  |       |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
|          |        | (ii)   |        | DNA                                                                                                                                                                                                                          |  |  |  | 1               |        |        |        |        |   | 1     |  |       |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
|          | (b)    | (i)    |        | 3 : 1                                                                                                                                                                                                                        |  |  |  |                 |        | 1      |        |        |   | 1     |  | 1     |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
|          |        | (ii)   | I      | <u>Heterozygous</u>                                                                                                                                                                                                          |  |  |  |                 |        | 1      |        |        |   | 1     |  |       |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
|          |        |        | II     | <u>homozygous recessive</u>                                                                                                                                                                                                  |  |  |  |                 |        | 1      |        |        |   | 1     |  |       |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
|          | (c)    | (i)    |        | <table><tr><td>sex</td><td>X only</td><td>Y only</td><td>X or Y</td></tr><tr><td>female</td><td>✓</td><td></td><td></td></tr><tr><td>male</td><td></td><td></td><td>✓</td></tr></table><br>one mark for each correct row (2) |  |  |  | sex             | X only | Y only | X or Y | female | ✓ |       |  | male  |  |      | ✓ | 2 |  |  |  |  |  | 2 |  |  |  |  |  |  |  |
| sex      | X only | Y only | X or Y |                                                                                                                                                                                                                              |  |  |  |                 |        |        |        |        |   |       |  |       |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
| female   | ✓      |        |        |                                                                                                                                                                                                                              |  |  |  |                 |        |        |        |        |   |       |  |       |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
| male     |        |        | ✓      |                                                                                                                                                                                                                              |  |  |  |                 |        |        |        |        |   |       |  |       |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
|          |        | (ii)   |        | ½, half, 50/50, 0.5, 50%                                                                                                                                                                                                     |  |  |  |                 |        | 1      |        |        |   | 1     |  |       |  |      |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |
|          |        |        |        | Question 4 total                                                                                                                                                                                                             |  |  |  | 4               |        | 4      |        | 0      |   | 8     |  | 1     |  | 0    |   |   |  |  |  |  |  |   |  |  |  |  |  |  |  |